Mohid Farhan

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Presidential Gold Medal-winning physicist with research experience spanning particle physics, cosmology, dark matter phenomenology, and computational physics. Primary author of peer-reviewed publications in Physica Scripta, an accepted solo-authored paper in European Physical Journal C, with several manuscripts under review at JCAP, EPJC, MNRAS, and Chinese Physics C, and an invited peer reviewer for Physica D: Nonlinear Phenomena (Elsevier). Recognized with competitive international scholarships for graduate research. Expertise in theoretical modelling, dark-sector phenomenology, and scientific computing.

Education

Sep 2021 – Sep 2025 — **BS Physics, Institute of Space Technology**
Presidential Gold Medal (Top Physics Graduate, Class of 2025) with a CGPA of 3.84/4, graduating magna cum laude. Research focused on Particle Physics, Cosmology, and Quantum Field Theory.

Scholarly Works

Published

- Naeem, M. and **Farhan, M.**, "Layered Dark Structure with a Structuring Field: A Z_4-Symmetric Inert Doublet-Singlet Realization and Implications for the S_8 Tension." *Physica Scripta* **(2026)**.
  Official: https://doi.org/10.1088/1402-4896/ae4e6e | Preprint: https://arxiv.org/abs/2603.27096
- **Farhan, M.**, "Symmetry-Protected Momentum Exchange between Dark Matter and Dark Energy." *Physica Scripta* **(2026)**.
  Official: https://doi.org/10.1088/1402-4896/ae6b13 | Preprint: https://arxiv.org/abs/2603.07879
- **Farhan, M.**, "The Impossible Triangle: A No-Go for Symmetry-Protected Scalar Portals in Interacting Dark Energy." *Journal of Cosmology and Astroparticle Physics* **(2026)**.
  Accepted. Preprint: https://arxiv.org/abs/2607.10859

Under Review

- **Farhan, M.**, Hassan, I., Usman, M., and Tahir, N., "Dark Matter and Dark Energy in Three-Higgs Doublet Model." Submitted to *Chinese Physics C (CPC)*.
  Preprint: https://arxiv.org/abs/2512.09404
- **Farhan, M.**, "Relic Density Topology as a Discriminatory Tool: A Comparative Analysis of IDM, MSSM, and NMSSM Dark Matter." Submitted to *European Physical Journal C (EPJC)*.
  Preprint: https://arxiv.org/abs/2512.09072 | **Presented at the IEEE-ICAST Conference.**
- **Farhan, M.**, "Discriminating dark-sector self-interaction regimes through cosmological observables." Submitted to *Monthly Notices of the Royal Astronomical Society (MNRAS)*.
  Preprint: https://doi.org/10.5281/zenodo.21376362
- Mohammad, D., Usman, M., and **Farhan, M.**, "Low-Redshift CPL fits in a Coupled Inflaton - Dark Energy Model" Submitted to *Physica Scripta*.
  Preprint: Available on Request.

Pedagogical Reports

- **Farhan, M.**, "Lecture Notes on Cosmological Perturbation Theory: The Fluid Equations for the Dark Sector." *Zenodo Pedagogical Report* **(2026)**.
  Official: https://doi.org/10.5281/zenodo.19535958
- **Farhan, M.**, "Lecture Notes on Quantum Electrodynamics: The Birth of Electromagnetism from Gauge Theory." *Zenodo Pedagogical Report* **(2026)**.
  Official: https://doi.org/10.5281/zenodo.22729433

Research Experience

Oct 2025 – Present

Independent Researcher, Multi-Component Dark Sector Frameworks.
Helped develop the **LDS-SF** framework to resolve the S_8 tension through scale-dependent growth derived from dark sector perturbation matrices. Provided the cosmological framework with a particle physics foundation by implementing a Z_4-symmetric Inert Doublet Singlet Model (**Z_4-IDSM**) in C-based software like **CLASS** and **micrOMEGAs**, mapping microscopic interactions to an effective sound speed (c_s^2). Derived upto two-loop Renormalization Group Equations (RGEs) for singlet mass parameters to ensure the radiative stability of the dark energy scale, and mapped it's cosmological effects in **CLASS** to study **Momentum Exchange Dark Matter-Dark Energy Interactions**.

Jul 2024 – Present

Theoretical Physics Researcher, Final Year Project and Independent Extension.
Led an advanced theoretical investigation into **dark matter and dark energy unification frameworks**, exploring extensions of the Standard Model such as the Three-Higgs Doublet Model (3HDM) and the Inert Doublet plus Singlet Model (IDSM). Utilized computational and symbolic tools including **SARAH**, **SPheno**, **micrOMEGAs**, **CalcRGE**, and **FeynCalc** to compute observables, relic densities, and parameter dependencies. Conducted comparative computational analysis of dark matter relic density topology across IDM, MSSM, and NMSSM.

Honors and Awards

2026

- **University of Queensland Graduate School Scholarship (UQGSS / RTP)**. Full scholarship for Master of Philosophy, including living stipend (AUD $39,200/year), full tuition fee offset, and Overseas Student Health Cover (OSHC). Competitive international merit-based award. (QS World Ranking 2026: 40)
- **King Fahd University of Petroleum and Minerals Direct PhD Awardee**, awarded a fully funded PhD Scholarship at KFUPM, including 10000 SAR/year stipend (extendable), air-conditioned and furnished housing, full tuition fee offset, annually reimbursed air tickets, and subsidized meals (QS World Ranking 2026: 63).
- **Ad-hoc Reviewer**, *Physica D: Nonlinear Phenomena* (Elsevier). Invited referee for a manuscript on dark-energy reconstruction and multi-field dark-sector models, evaluating cosmological perturbation theory and dark matter-dark energy interactions.

2025

- **IST President's Gold Medalist in Physics 2025**, Highest academic honor awarded to the student with the best GPA of the cohort.
- **Merit-based scholarships** , Awarded **6 times** due to obtaining the highest GPA of the batch for the semester (from 2022 to 2025).
- **Magna cum laude**, Awarded to high-performing students with a final CGPA of 3.7 or more.
- **Invited Speaker**, 18th National Symposium on Frontiers in Physics.
- **Best Final Year Project for Cosmology**, Single-handedly represented cosmology in the External Selection Committee (ESC) evaluation for the best FYP.
- **Adjudicator at World Space Week 2025**, Selected as evaluator and judged an English Debating Competition for high-schoolers.
- **Author at International Conference on Applications of Science and Technology**, Invited as author of a single-author manuscript in ICAST.

2024

- **Participant in International School on Physics & Allied Disciplines**, ICTP & National Center for Physics.

2023

- **Laptop Awarded**, by merit-based selection under the Prime Minister Laptop Scheme Program.

Academic Service

Sep 2026 — **Ad-hoc Reviewer,** Physica D: Nonlinear Phenomena (Elsevier).
Refereed a full-length manuscript on dark-energy potential reconstruction and multi-field dark-sector dynamics. Evaluation focused on cosmological perturbation theory, dark matter-dark energy interactions, interacting dark energy, and self-interacting dark matter frameworks. Reviewed the mathematical formulation of the inverse reconstruction problem, consistency conditions for single-field vs multi-field realizations, numerical verification of analytical predictions, and connections to observational degeneracies in the dark sector.

Research-Based Internships and Conferences

Jul 2023 – Sep 2023 — **Internee,** National Centre for Physics (NCP).
Worked on matter-radiation interactions on a project titled **Shielding Against Cosmic Rays and Solar Energetic Particles for Space Crafts**. Analyzed particle flux models, attenuation behavior of shielding materials, and CMB photon interaction limits relevant to long-duration missions. Contributed to simulations estimating dose deposition and evaluating mitigation strategies under varying orbital and interplanetary conditions.

Jul 2024 – Sep 2024 — **Internee,** National Center for GIS and Space Applications.
Acquired hands-on experience in **astronomical data acquisition and analysis**, focusing on telescopic observations and image processing. Worked with BASS Project, Registax, PIPP, Siril, Stellarium, AstroImageJ, N.I.N.A and Astroplanner for calibration, reduction, and analysis of stellar and planetary data. Performed spectroscopic and photometric measurements to extract stellar parameters such as temperature and composition.

Nov 2025 — **Author and Presenter,** IEEE-ICAST Conference.
Presented original research titled *"Relic Density Topology as a Discriminatory Tool: A Comparative Analysis of IDM, MSSM, and NMSSM Dark Matter."* Developed a topological mapping approach to distinguish between SUSY and non-SUSY dark matter candidates based on thermal relic abundance.

Oct 2025 — **Adjudicator,** World Space Week (WSW).
Selected as a technical evaluator for the Scientific Debating Competition, assessing scientific arguments and communication clarity for high-school participants under the WSW global initiative.

Feb 2025 — **Invited Speaker,** 18th National Symposium on Frontiers in Physics.
Delivered a technical talk at the **National Centre for Physics (NCP)** symposium on *"Unified Frameworks for Dark Matter and Dark Energy,"* presenting results from the 3HDM and IDSM phenomenological scans. Engaged with the national physics community on the implications of multi-Higgs models for cosmological tensions.

Dec 2024 — **Co-organizer and Participant,** 2nd Workshop on General Relativity, Cosmology, Astrophysics and Particle Physics 2024.
Attended and helped organize the second edition, hosted by National Centre for GIS and Space Application and Institute of Space Technology. Had discussions with local senior scientists on Virial clouds as dark matter candidates.

Jun 2024 — **Participant,** International School on Physics & Allied Disciplines (ISPAD).
Attended the intensive school hosted by the **ICTP** and the **National Centre for Physics (NCP)**, focusing on advanced topics in Particle Physics and High Energy Physics.

Dec 2024 — **Co-organizer and Participant,** 1st Workshop on Cosmology.
Attended and helped organize the event hosted the inaugural workshop, along with the research group, led by my supervisor, in an attempt to bring a workshop to our university (Institute of Space Technology).

Science Communication and Outreach

July 2025 – Present — **Mathematics and Geography Teacher,** The City School Educational Services. Designing and delivering engaging lectures in Mathematics and Geography for middle and high school levels. Implementing conceptual and inquiry-based learning approaches aligned with The City School's curriculum. Fostering critical thinking, problem-solving, and interdisciplinary understanding among the next generation of learners.

Jul 2024 – Present — **Science Communication and Blogging,** https://mohidfarhan.github.io/blog/. Authoring explanatory essays and visual content that make complex physics and cosmology concepts accessible to broader audiences. Using **Manim** and **Python** to illustrate key ideas in particle physics, nuclear physics, quantum field theory, dark matter, and general relativity.

Jul 2022 – Present — **World Space Week,** National Center for GIS and Space Applications. Engaged in public science outreach as module designer, team lead, and adjudicator across multiple editions of World Space Week, promoting space science education and engagement.

Skills

New Physics Research Tools — **micrOMEGAs, SPheno, SARAH, CalcRGE, CLASS**: Development and implementation of Beyond the Standard Model (BSM) frameworks, with computation of dark matter related observables, mass spectra, CMB predictions and lensing data.

Astronomy and Big Data Tools — **AstroImageJ, BASS Project, Siril, PIPP, Registax, Stellarium, N.I.N.A.**: Data acquisition, calibration, and reduction of telescopic and CCD images for stellar and galactic photometry and spectroscopy. Experience with extraction from large data sets and surveys, of determining stellar parameters and construction of light curves for eclipsing binary stars.

Computational Tools — **Mathematica, Python, C++**: Symbolic and numerical computation, automation of model analysis, and data visualization.

Science Communication Tools — **LaTeX, GitHub, Anaconda, Manim**: Scientific writing and version control, environment management, and production of educational animations for physics outreach.

Analytical and Interpersonal — **Advanced problem-solving** and **critical thinking** developed through theoretical model design and independent research. **Scientific communication** demonstrated through talks, publications, and outreach. **Leadership and collaboration** strengthened through research and educational initiatives.

References

Available on request.